

Xiaoying Liao

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Education

Johns Hopkins University

2024–2025

M.S.E. in Biomedical Engineering; GPA: 4.00/4.00; advisor: Prof. Carlo Iomini.

Courses: Cellular Engineering; Tissue Engineering; Cell & Tissue Engineering Lab; Regenerative and Immune Engineering; Human Genetics; Systems Pharmacology and Personalized Medicine.

University of Sydney

2020–2024

B.Sc. (Honours) in Statistics and Economics; GPA: 3.90/4.00; rank: 1/200; advisor: Prof. Jean Yang.

Courses: Probability and Estimation Theory; Statistical Inference; Stochastic Process; Advanced Statistical Modelling; Statistical Machine Learning; Life and Evolution.

Research Experience

Stanford University — Research Assistant

2026–Present

Supervisor: Prof. Xiaojie Qiu

- Analyze PCW7–8 human fetal heart spatial transcriptomics to characterize cardiac cell states, spatial niches, cell–cell communication, and 3D tissue architecture; build an E5.5–E7.5 mouse embryogenesis atlas.
- Lead a pediatric cardiomyopathy study integrating single-cell multi-omics and spatial transcriptomics across VAD support and recovery outcomes to identify myocardial recovery programs and potential biomarkers.

Johns Hopkins University — Research Assistant

2024–2026

Supervisor: Prof. Carlo Iomini, Wilmer Eye Institute

- Investigated Hedgehog-responsive progenitors and Ift88–Smo genetic interactions in meibomian gland development, epithelial proliferation, meibocyte maturation, and lipid homeostasis.
- Generated and analyzed P5/P10 meibomian gland scRNA-seq to characterize developmental cell states and pathways associated with gland maturation and Hedgehog signaling.

University of Pittsburgh — Research Collaboration

2025–2026

Collaborators: Dr. Huanhuan Xiao and Prof. Liyun Zhang

- Developed R workflows to quantify photopic/scotopic visual motor responses in pde6b mutant zebrafish; compared genotypes and RA, 17-AAG, and DHA treatments and generated publication figures.

University of Sydney — Research Assistant

2023–2024

Supervisors: Prof. Jean Yang and Prof. Clara Grazian

- Preprocessed spatial transcriptomics in Python/R and benchmarked K-means, Seurat, BayesSpace, and STitch3D across simulated and real datasets to assess spatial domains and method performance.

Programming & Research Skills

Programming & data analysis: Python, R, SQL, MATLAB, Linux, Git, PyTorch, scikit-learn, NumPy, pandas, Scanpy, Seurat, Bioconductor, Cell Ranger, GraphPad Prism, FIJI/ImageJ.

Computational biology: Single-cell/spatial transcriptomics; statistical modeling; machine/deep learning; biological foundation models; multimodal/multi-omics integration; perturbation/drug-response modeling; spatial omics benchmarking.

Experimental biology: 10x Genomics single-cell sequencing; Illumina spatial transcriptomics; cell/tissue culture; immunostaining; confocal/two-photon microscopy; PCR/qRT-PCR; Western blotting; genotyping; DNA/RNA extraction; mouse procedures.

Professional Experience

JD.com — Data Analyst Intern

Feb.–May 2022; Beijing, China

Hive SQL retail data management, time-series forecasting, and regression; R/Python customer segmentation associated with 25% quarterly sales growth; Python dashboards associated with 60% higher operational efficiency.

ByteDance–TikTok — Data Operation Intern

Dec. 2021–Feb. 2022; Beijing, China

CRM market research; SQL/Python KPI monitoring and BI dashboards; AIPL user segmentation and targeted strategies associated with a 15% increase in brand advertising click-through rate.

Teaching Experience

Johns Hopkins University — Teaching Assistant (2025–2026): Cell & Tissue Engineering Lab: cell culture, gene delivery, metabolic glycoengineering, tissue modeling, data analysis, and scientific reports.

University of Sydney — Teaching Assistant (2023–2024): STAT2011 Probability and Estimation Theory; DATA2002 Data Analytics: Learning from Data; probability, R programming, analytics, and machine learning workshops.

Published Papers, Preprints & Posters

1. Python Song and **Xiaoying Liao**. *ProactiveEngine: Proactive Learning with Lifelong Memory via World Models*. Empirical Methods in Natural Language Processing (EMNLP, Main Conference), 2026.
2. Miao Cui, Cinlong Huang, **Xiaoying Liao**, Xiaoke Song, Stephanie Conley, Jesse Engreitz, and Xiaojie Qiu. *Spatio-temporal Transcriptomic Profiling of Serial Human Fetal Heart Sections Reveals Cellular Architecture and Functional Niches during development*. Stanford Genetics Department Retreat (poster), 2026.
3. Huanhuan Xiao, **Xiaoying Liao**, Céline Portal, Zhenzhen Pan, Eileen Chen, Igor A. Butovich, and Carlo Iomini. *Hedgehog Signaling Maintains Meibomian Gland Progenitors and Lipid Homeostasis with Implications for Meibomian Gland Dysfunction Therapeutics*. Investigative Ophthalmology & Visual Science, 67(7):3883, 2026.
4. Nianping Liu, Xuanzhi Chen, Miao Cui, **Xiaoying Liao**, Xiaoke Song, Sairam Pantham, Weize Xu, and Xiaojie Qiu. *Bio-Babel: Autonomous Cross-Language Reconstruction of Agent-Ready Computational Biology Ecosystems*. bioRxiv, 2026.
5. Tianyang Wang, Xi Xiao, and **Xiaoying Liao**. *Foundation Models for Radiology Report Generation*. One chapter in medical field's book, 2026; Under review.
6. Xi Xiao, Yunbei Zhang, Lin Zhao, Yiyang Liu, **Xiaoying Liao**, Zheda Mai, Xingjian Li, Xiao Wang, Hao Xu, Jihun Hamm, Xue Lin, Min Xu, Qifan Wang, Tianyang Wang, and Cheng Han. *Prompt-based Adaptation in Large-scale Vision Models: A Survey*. Transactions on Machine Learning Research, 2026.
7. Junjie Li, Xi Xiao, Yunbei Zhang, Chen Liu, Lin Zhao, **Xiaoying Liao**, Yingrui Ji, Janet Wang, Jianyang Gu, Yingqiang Ge, Weijie Xu, Xi Fang, Xiang Xu, Tianchen Zhao, Youngeun Kim, Tianyang Wang, Jihun Hamm, Smita Krishnaswamy, Jun Huan, and Chandan K. Reddy. *Agent Harness Engineering: A Survey*. Under review at ACL, 2026.
8. Tianyang Wang, Xi Xiao, Gaofer Chen, **Xiaoying Liao**, Guo Cheng, and Yingrui Ji. *Boosting Active Learning with Knowledge Transfer*. 2025 IEEE 37th International Conference on Tools with Artificial Intelligence, 2025; pp. 1124–1131.

Manuscripts in Preparation

1. Huanhuan Xiao, Xiangqian Shi, Abi Cha, Emma Liu, Randi Marshall, **Xiaoying Liao**, Molly Voit, Uzoamaka Nwagbo, Frazer Matthews, James H. Thierer, Meera Saxena, Ximin Li, Tae-In Kam, Xiangrong Kong, Jeff S. Mumm, Liyun Zhang. *Novel zebrafish genetic models of pde6b mutants exhibit rod photoreceptor degeneration*. Completed; planned submission to Communications Biology by October 2026.
2. Huanhuan Xiao, **Xiaoying Liao**, Céline Portal, Zhenzhen Pan, Eileen Chen, Igor A. Butovich, Carlo Iomini. *Genetic interaction between primary cilia and Hedgehog signaling regulates meibomian gland development*. In preparation; expected submission by December 2026.
3. Huanhuan Xiao, Liyun Zhang, **Xiaoying Liao**, Emma Liu, Shaojun Du. *Expression and functional analysis of Hspa8 in early zebrafish development*. Completed; planned submission to Cells in 2026.

Honors, Awards & Certifications

Roland Berger Case Competition, 2nd prize (2/2000), 2023; 3rd Meituan Business Analysis Elite Competition participation certificate, 2023; Huawei Financial Elite Challenge, Australia, 3rd prize (3/1000), 2021.

China National High School Biology Olympiad, 2nd prize, 2018; Dalyell Scholar, 2022; Vice Chancellor's Global Mobility Scholarship, 2020.

Certified Financial Risk Manager (FRM), Part I and Part II, 2023.

Academic Service & Conference Participation

Conference reviewer: AAAI (2025–2026); ICLR (2027).

2026: Stanford Genetics Department Retreat; ARVO; Johns Hopkins Cell Biology Symposium; Annual Postdoctoral Research Meeting. 2025: Maryland Stem Cell Colloquium; Annual Wilmer Research Meeting.

2023 (University of Sydney): Sydney Bioinformatics Research Symposium; Single-Cell Plus – Data Science Challenges in Single-Cell Research; Unlocking Single Cell Spatial Omics Analyses with Scdny.

Additional Research Projects

Gne knockout validation, Johns Hopkins (2024): Led a CHO-K1 CRISPR validation project using PCR, Sanger sequencing, and Western blotting.

Single-cell benchmarking evaluation, Sydney (2023): Developed a 70+ question evaluation framework; screened 245 studies from 563; applied linear/logistic regression to characterize benchmarking practices.

Type 2 diabetes prediction, Sydney (2022): Led a machine learning capstone using linear/ridge/Lasso regression and cross-validated decision trees to model HbA1c and inform dietary recommendations.